A Comprehensive Analytical Framework for Bio-Inspired Metaheuristic Optimization Techniques in Solving Multi-Objective Optimal Power Flow Problems with FACTS Device Placement and AI-Enhanced Control Strategies Across Standard IEEE Test Systems with Varying Complexity and Renewable Energy Integration

Author Name[1], Co-Author Name[2], Co-Author Name[3]

[1]*Department of Electrical Engineering, University Name, City, Country*
[2]*Power Systems Research Center, University Name, City, Country*
[3]*Department of Computer Science, University Name, City, Country*

Email: author@university.edu, coauthor@university.edu, coauthor2@university.edu

Abstract

This paper presents a systematic investigation of advanced metaheuristic optimization techniques for solving Optimal Power Flow (OPF) problems across standard IEEE test systems with integrated Flexible AC Transmission Systems (FACTS) devices and Artificial Intelligence (AI) methodologies. The research examines five prominent bio-inspired algorithms—Genetic Algorithm (GA), Ant Colony Optimization (ACO), Particle Swarm Optimization (PSO), Firefly Algorithm (FA), and Bacterial Foraging Optimization (BFO)—applied to IEEE 9-bus, 14-bus, 30-bus, 69-bus, and 118-bus test systems. The study focuses on optimizing multiple objectives including generation cost minimization, transmission loss reduction, voltage profile improvement, and system stability enhancement while integrating various FACTS controllers (SVC, TCSC, UPFC, and DPFC) to address the challenges posed by renewable energy penetration. The research develops novel hybrid AI-enhanced optimization frameworks that combine the exploratory advantages of metaheuristic algorithms with the predictive capabilities of machine learning and deep learning techniques to improve convergence characteristics and solution quality. Extensive comparative analyses across test systems of increasing complexity provide insights into algorithm performance, computational efficiency, and effectiveness of FACTS placement strategies under different operating conditions. The findings demonstrate significant improvements in power system performance metrics while establishing practical guidelines for optimal deployment of FACTS devices using AI-integrated optimization approaches in modern power systems with high renewable energy penetration.

Keywords*—Bio-inspired optimization, genetic algorithm, particle swarm optimization, ant colony optimization, firefly algorithm, bacterial foraging optimization, optimal power flow, FACTS devices, multi-objective optimization, artificial intelligence, renewable energy integration.*

I. INTRODUCTION

A. Background and Motivation

Modern power systems face unprecedented challenges due to increasing renewable energy integration, market deregulation, and growing demand complexity. These challenges necessitate sophisticated optimization approaches for power system operation and planning. Optimal Power Flow (OPF), a fundamental optimization problem in power systems, aims to determine the optimal settings of control variables that minimize specified objective functions while satisfying various operational constraints. The complexity of OPF problems has grown substantially with the integration of renewable energy sources, which introduce variability and uncertainty into the system.

Flexible AC Transmission Systems (FACTS) devices have emerged as critical tools for enhancing power system controllability, stability, and transmission capacity. These devices, including Static VAR Compensators (SVC), Thyristor Controlled Series Capacitors (TCSC), Unified Power Flow Controllers (UPFC), and Distributed Power Flow Controllers (DPFC), offer dynamic control of various power system parameters. However, determining the optimal location, sizing, and control settings of these devices presents a complex optimization challenge that conventional mathematical techniques struggle to address effectively.

The limitations of traditional optimization methods have spurred interest in metaheuristic optimization algorithms, which offer robust solutions to complex, non-linear, and multi-objective problems. Concurrently, the rapid advancement of Artificial Intelligence (AI) techniques provides unprecedented opportunities to enhance these optimization processes through improved prediction, adaptive learning, and intelligent decision-making. This paper explores the integration of these advanced approaches to address the multi-faceted challenges of modern power systems.

B. Research Objectives and Scope

The primary objectives of this research are:

1) To develop a comprehensive analytical framework for evaluating the performance of five prominent bio-inspired metaheuristic optimization techniques (GA, ACO, PSO, FA, and BFO) in solving multi-objective OPF problems.

2) To formulate mathematical models for various FACTS devices and integrate them into the OPF problem formulation.

3) To design novel hybrid optimization approaches that combine the strengths of metaheuristic algorithms with AI techniques.

4) To conduct extensive comparative analyses of algorithm performance across IEEE test systems of different sizes and complexities.

5) To establish guidelines for the optimal placement and parameter setting of FACTS devices to enhance power system performance with high renewable energy penetration.

C. Paper Organization

The remainder of this paper is organized as follows: Section II provides a comprehensive literature review on metaheuristic optimization techniques and their applications in power systems. Section III presents the mathematical formulation of the multi-objective OPF problem and FACTS device models. Section IV describes the bio-inspired metaheuristic algorithms in detail. Section V introduces the proposed AI-enhanced optimization frameworks. Section VI outlines the test systems and simulation methodology. Section VII presents and discusses the results of the comparative analyses. Finally, Section VIII concludes the paper with key findings and future research directions.

II. LITERATURE REVIEW

A. Evolution of Optimal Power Flow Techniques

The OPF problem was first introduced by Carpentier in 1962 [1] and has since evolved significantly. Early approaches utilized classical optimization techniques such as linear programming, non-linear programming, quadratic programming, and mixed-integer programming [2]. While effective for specific problems, these methods often struggled with the non-linear, non-convex nature of practical OPF problems, especially those involving multiple objectives and constraints.

With the increasing complexity of power systems, researchers began exploring heuristic and metaheuristic approaches. Dommel and Tinney [3] applied the Newton-Raphson method to solve the OPF problem, setting the foundation for numerous subsequent studies. Interior point methods, as demonstrated by Granville [4], improved computational efficiency but still faced challenges with local optimality and non-convexity. The integration of renewable energy sources and FACTS devices has further complicated the OPF problem, necessitating more sophisticated optimization approaches.

B. Bio-Inspired Metaheuristic Optimization Techniques

Bio-inspired metaheuristic algorithms have gained prominence in power system optimization due to their ability to handle complex, non-linear, and multi-objective problems. The Genetic Algorithm (GA), inspired by natural evolution, was among the first to be applied to power system optimization problems [5]. Wong and Wong [6] demonstrated GA's effectiveness for economic dispatch problems, while Miranda et al. [7] applied it to reactive power planning.

Particle Swarm Optimization (PSO), inspired by the social behavior of bird flocking, was introduced by Kennedy and Eberhart [8] and has been widely applied to power system optimization. Abido [9] utilized PSO for multi-objective OPF, demonstrating its effectiveness in generating Pareto-optimal solutions. Ant Colony Optimization (ACO), based on the foraging behavior of ants, has been applied to various power system optimization problems, including optimal placement of FACTS devices [10].

More recent bio-inspired algorithms such as the Firefly Algorithm (FA) [11] and Bacterial Foraging Optimization (BFO) [12] have shown promising results in solving complex power system optimization problems. These algorithms offer diverse mechanisms for exploring the solution space and escaping local optima, making them suitable for multi-objective optimization problems in modern power systems.

C. FACTS Devices in Power System Optimization

FACTS devices have become integral components of modern power systems, providing enhanced controllability and stability. Hingorani and Gyugyi [13] provided a comprehensive overview of FACTS technologies and their applications. The optimal placement and sizing of these devices have been extensively studied. Gerbex et al. [14] applied genetic algorithms to optimize the location of multiple FACTS devices, while Saravanan et al. [15] utilized PSO for the same purpose.

The integration of FACTS devices into OPF formulations has been addressed by several researchers. Zhang et al. [16] developed a comprehensive OPF model including SVC and TCSC, while Cai et al. [17] incorporated UPFC into the OPF formulation. These studies demonstrated the significant impact of FACTS devices on power system performance, including reduced transmission losses, improved voltage profiles, and enhanced stability margins.

However, the complexity introduced by FACTS devices in OPF problems necessitates advanced optimization techniques. Shaheen et al. [18] compared various evolutionary algorithms for solving OPF problems with FACTS devices, highlighting the need for robust and efficient optimization approaches.

D. AI Integration in Power System Optimization

The integration of Artificial Intelligence (AI) techniques with metaheuristic optimization has emerged as a promising research direction. Machine learning approaches, particularly neural networks and fuzzy logic, have been applied to enhance the performance of optimization algorithms. Krami et al. [19] developed a neural network-based approach for predicting optimal FACTS device parameters, reducing computational complexity.

Deep learning techniques have further extended the capabilities of AI in power system optimization. Wang et al. [20] utilized deep reinforcement learning for dynamic optimization of FACTS device settings, demonstrating improved adaptability to changing system conditions. Hybrid approaches combining metaheuristic algorithms with AI techniques have shown particular promise. Li et al. [21] integrated machine learning with PSO to enhance convergence and solution quality for complex OPF problems.

Despite these advancements, there remains a gap in the literature regarding comprehensive analytical frameworks that systematically evaluate and enhance bio-inspired optimization techniques for multi-objective OPF problems with FACTS devices, particularly in the context of high renewable energy penetration. This paper addresses this gap by developing novel hybrid approaches and conducting extensive comparative analyses across different test systems.

III. MATHEMATICAL FORMULATION

A. Multi-Objective Optimal Power Flow Problem

The multi-objective OPF problem aims to simultaneously optimize multiple, often conflicting, objective functions subject to various equality and inequality constraints. Mathematically, it can be formulated as:

$$\min \mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})]$$

Subject to: (1)

$$\mathbf{g}(\mathbf{x}) = 0$$
$$\mathbf{h}(\mathbf{x}) \le 0 \qquad (2)$$

where $\mathbf{F}(\mathbf{x})$ is the vector of objective functions, $\mathbf{x}$(3) is the vector of control and state variables, $\mathbf{g}(\mathbf{x})$ represents the equality constraints (power flow equations), and $\mathbf{h}(\mathbf{x})$ represents the inequality constraints (operational and security constraints).

In this study, we consider the following objective functions:

1) Minimization of Generation Cost:

$$f_1(\mathbf{x}) = \sum_{i=1}^{N_G}(a_i + b_iP_{Gi} + c_iP_{Gi}^2)$$

where N_G is the number of generators, P_{Gi} is the(4) active power output of generator i, and a_i, b_i, and c_i are the cost coefficients of generator i.

2) Minimization of Transmission Losses:

$$f_2(\mathbf{x}) = \sum_{k=1}^{N_L} g_k\left[V_i^2 + V_j^2 - 2V_iV_j\cos(\delta_i - \delta_j)\right]$$

where N_L is the number of transmission lines, g_k(5) is the conductance of line k connecting buses i and j, V_i and V_j are the voltage magnitudes at buses i and j, and δ_i and δ_j are the voltage angles at buses i and j.

3) Voltage Profile Improvement:

$$f_3(\mathbf{x}) = \sum_{i=1}^{N_B}(V_i - V_i^{ref})^2$$

where N_B is the number of buses, V_i is the(6) voltage magnitude at bus i, and V_i^{ref} is the reference (desired) voltage magnitude at bus i.

4) System Stability Enhancement:

$$f_4(\mathbf{x}) = \sum_{i=1}^{N_G}\sum_{j=1}^{N_L} S_{ij}^2$$

where S_{ij} is the sensitivity factor representing the(7) impact of generator i on the stability of line j.

The constraints of the OPF problem include:

1) Power Flow Equations (Equality Constraints):

$$P_i = V_i\sum_{j=1}^{N_B}V_j(G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij})$$
$$Q_i = V_i\sum_{j=1}^{N_B}V_j(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}) \qquad (8)$$

where P_i and Q_i are the active and reactive power(9) injections at bus i, G_{ij} and B_{ij} are the real and imaginary parts of the element in the bus admittance matrix corresponding to buses i and j, and $\delta_{ij} = \delta_i - \delta_j$ is the voltage angle difference between buses i and j.

2) Generation Limits (Inequality Constraints):

$$P_{Gi}^{min} \le P_{Gi} \le P_{Gi}^{max}, \quad i = 1, 2, \ldots, N_G$$
$$Q_{Gi}^{min} \le Q_{Gi} \le Q_{Gi}^{max}, \quad i = 1, 2, \ldots, N_G \qquad (10)$$

3) Voltage Limits (Inequality Constraints): (11)

$$V_i^{min} \le V_i \le V_i^{max}, \quad i = 1, 2, \ldots, N_B$$

4) Line Flow Limits (Inequality Constraints): (12)

$$|S_l| \le S_l^{max}, \quad l = 1, 2, \ldots, N_L$$

where $|S_l|$ is the apparent power flow in line l(13) and S_l^{max} is the thermal limit of line l.

B. FACTS Device Models

1) Static VAR Compensator (SVC):

The SVC is modeled as a variable shunt susceptance B_{SVC} at bus i. The power injection model of SVC can be represented as:

$$P_{SVC} = 0$$
$$Q_{SVC} = -V_i^2B_{SVC}$$
$$B_{SVC}^{min} \le B_{SVC} \le B_{SVC}^{max} \qquad (14)$$

where B_{SVC}^{min} and B_{SVC}^{max} are the minimum and(15) maximum susceptance limits of the SVC.

2) Thyristor Controlled Series Capacitor (TCSC):

The TCSC is modeled as a variable reactance X_{TCSC} connected in series with the transmission line between buses i and j. The modified reactance of the line is:

$$X_{ij} = X_{line} + X_{TCSC}$$
$$X_{TCSC}^{min} \le X_{TCSC} \le X_{TCSC}^{max} \qquad (16)$$

where X_{line} is the original reactance of the line,(17) and X_{TCSC}^{min} and X_{TCSC}^{max} are the minimum and maximum reactance limits of the TCSC.

3) Unified Power Flow Controller (UPFC):

The UPFC is modeled as a combination of a shunt voltage source and a series voltage source. The power injection model of UPFC can be represented as:

$$P_i^{UPFC} = -V_iV_jB_{ij}\sin(\delta_i - \delta_j - \gamma_{ij})$$
$$Q_i^{UPFC} = -V_i^2B_{ij} - V_iV_jB_{ij}\cos(\delta_i - \delta_j - \gamma_{ij})$$
$$P_j^{UPFC} = V_iV_jB_{ij}\sin(\delta_i - \delta_j - \gamma_{ij})$$
$$Q_j^{UPFC} = -V_j^2B_{ij} + V_iV_jB_{ij}\cos(\delta_i - \delta_j - \gamma_{ij})$$

where B_{ij} is the susceptance of the line(18) connecting buses i and j, and γ_{ij} is the phase angle of the series voltage source.

4) Distributed Power Flow Controller (DPFC):

The DPFC is an advanced FACTS device derived from the UPFC. It utilizes multiple small series converters instead of a large one, eliminating the common DC link. The power injection model of DPFC can be represented as:

$$P_i^{DPFC} = -V_iV_jB_{ij}\sin(\delta_i - \delta_j - \gamma_{ij}) + V_i^2G_{sh}$$
$$Q_i^{DPFC} = -V_i^2B_{ij} - V_iV_jB_{ij}\cos(\delta_i - \delta_j - \gamma_{ij}) - V_i^2B_{sh}$$
$$P_j^{DPFC} = V_iV_jB_{ij}\sin(\delta_i - \delta_j - \gamma_{ij})$$
$$Q_j^{DPFC} = -V_j^2B_{ij} + V_iV_jB_{ij}\cos(\delta_i - \delta_j - \gamma_{ij})$$

where G_{sh} and B_{sh} are the conductance and(19) susceptance of the shunt converter.

C. Multi-Objective Optimization Approaches

In this paper, we employ two approaches to handle the multi-objective optimization problem:

1) Weighted Sum Method: This method converts the multi-objective problem into a single-objective problem by assigning weights to each objective function:

$$F(\mathbf{x}) = \sum_{i=1}^{m} w_if_i(\mathbf{x})$$

where w_i is the weight assigned to objective(20) function $f_i(\mathbf{x})$, and $\sum_{i=1}^m w_i = 1$.

2) Pareto Optimization: This approach aims to find the set of Pareto-optimal solutions, where no solution can be improved in one objective without degrading at least one other objective. A solution $\mathbf{x}_1$ dominates another solution $\mathbf{x}_2$ if:

a) $f_i(\mathbf{x}_1) \le f_i(\mathbf{x}_2)$ for all $i \in \{1, 2, \ldots, m\}$

b) $f_j(\mathbf{x}_1) < f_j(\mathbf{x}_2)$ for at least one $j \in \{1, 2, \ldots, m\}$

The set of all Pareto-optimal solutions forms the Pareto front, providing decision-makers with a range of trade-off solutions.

IV. BIO-INSPIRED METAHEURISTIC ALGORITHMS

A. Genetic Algorithm (GA)

The Genetic Algorithm is an evolutionary algorithm inspired by the process of natural selection. It operates on a population of potential solutions, applying genetic operators such as selection, crossover, and mutation to evolve better solutions over successive generations.

The basic steps of GA are as follows:

```
Algorithm 1: Genetic Algorithm
Initialize a population of N individuals (potential solutions)
Evaluate the fitness of each individual
while termination criterion not met do
    Select parents based on their fitness
    Generate offspring through crossover and mutation
    Evaluate the fitness of the offspring
    Replace the least fit individuals with the offspring
end while
return The best individual found
```

In the context of the OPF problem, each individual represents a vector of control variables, including generator outputs, voltage settings, transformer tap ratios, and FACTS device parameters. The fitness function is the weighted sum of the objective functions (for the weighted sum method) or the dominance rank (for Pareto optimization).

Genetic Algorithm Flowchart

Fig. 1. Flowchart of the Genetic Algorithm.

The mathematical formulation of GA can be represented as follows:

1) Selection: Parents are selected using methods such as roulette wheel selection, tournament selection, or rank-based selection. The probability of selection for an individual i using roulette wheel selection is:

$$p_i = \frac{f_i}{\sum_{j=1}^{N} f_j}$$

where f_i is the fitness of individual i and N is(21) the population size.

2) Crossover: Two parent individuals exchange genetic material to produce offspring. For a one-point crossover with crossover point c, the offspring are:

$$offspring_1 = [parent_1[1:c], parent_2[c+1:end]]$$
$$offspring_2 = [parent_2[1:c], parent_1[c+1:end]]$$

3) Mutation: Random changes are introduced to(22) maintain genetic diversity. For each gene in an individual, mutation occurs with probability p_m:

$$gene_i' = \begin{cases} gene_i + \Delta, & \text{with probability } p_m \\ gene_i, & \text{with probability } 1 - p_m \end{cases}$$

where Δ is a small random change within the(23) allowable range of the gene.

B. Particle Swarm Optimization (PSO)

Particle Swarm Optimization is a population-based stochastic optimization technique inspired by the social behavior of birds flocking or fish schooling. Each particle in the swarm represents a potential solution and moves through the search space based on its own experience and the experience of neighboring particles.

The basic steps of PSO are as follows:

```
Algorithm 2: Particle Swarm Optimization
Initialize a swarm of N particles with random positions and velocities
Evaluate the fitness of each particle
Set personal best (pbest) of each particle to its current position
Set global best (gbest) to the position of the best particle
while termination criterion not met do
    for each particle do
        Update velocity based on pbest and gbest
        Update position based on velocity
        Evaluate fitness
        Update pbest if current position is better
        Update gbest if current position is better than global best
    end for
end while
return The global best position found
```

The mathematical formulation of PSO can be represented as follows:

1) Velocity Update:

$$v_i^{t+1} = wv_i^t + c_1r_1(pbest_i - x_i^t) + c_2r_2(gbest - x_i^t)$$

2) Position Update: (24)

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

where v_i^t and x_i^t are the velocity and position of(25) particle i at iteration t, $pbest_i$ is the personal best position of particle i, $gbest$ is the global best position, w is the inertia weight, c_1 and c_2 are cognitive and social coefficients, and r_1 and r_2 are random numbers in the range $[0, 1]$.

Particle Swarm Optimization Flowchart

Fig. 2. Flowchart of the Particle Swarm Optimization algorithm.

C. Ant Colony Optimization (ACO)

Ant Colony Optimization is a probabilistic technique inspired by the foraging behavior of ants, which find the shortest path between their colony and a food source by depositing pheromones. In the context of the OPF problem, ACO can be adapted to continuous optimization by discretizing the search space.

The basic steps of ACO are as follows:

```
Algorithm 3: Ant Colony Optimization
Initialize pheromone trails
while termination criterion not met do
    Construct solutions for each ant based on pheromone trails
    Evaluate the fitness of each solution
    Update pheromone trails based on solution quality
    Apply pheromone evaporation
end while
return The best solution found
```

The mathematical formulation of ACO can be represented as follows:

1) Solution Construction: The probability of ant k choosing value j for variable i is:

$$p_{ij}^k = \frac{[\tau_{ij}]^\alpha \cdot [\eta_{ij}]^\beta}{\sum_{l=1}^{n_i}[\tau_{il}]^\alpha \cdot [\eta_{il}]^\beta}$$

where τ_{ij} is the pheromone level associated with(26) value j for variable i, η_{ij} is the heuristic information, α and β are parameters controlling the relative importance of pheromone versus heuristic information, and n_i is the number of possible values for variable i.

2) Pheromone Update: After all ants have constructed their solutions, the pheromone trails are updated:

$$\tau_{ij} = (1 - \rho)\cdot\tau_{ij} + \sum_{k=1}^{m}\Delta\tau_{ij}^k$$

where ρ is the pheromone evaporation rate, m is(27) the number of ants, and $\Delta\tau_{ij}^k$ is the amount of pheromone deposited by ant k:

$$\Delta\tau_{ij}^k = \begin{cases} 1/C^k, & \text{if ant } k \text{ uses value } j \text{ for variable } i \\ 0, & \text{otherwise} \end{cases}$$

where C^k is the cost of the solution constructed(28) by ant k.

D. Firefly Algorithm (FA)

The Firefly Algorithm is a metaheuristic optimization technique inspired by the flashing behavior of fireflies. Each firefly's brightness is proportional to its solution quality, and fireflies are attracted to brighter ones, enabling the swarm to converge towards the optimal solution.

The basic steps of FA are as follows:

```
Algorithm 4: Firefly Algorithm
Initialize a population of N fireflies with random positions
Define light intensity at each firefly based on the objective function
Define the absorption coefficient γ
while termination criterion not met do
    for each firefly i do
        for each firefly j do
            if intensity at j > intensity at i then
                Move firefly i towards j based on attractiveness
            end if
        end for
        Evaluate new solution and update intensity
    end for
    Rank fireflies and find the current best
end while
return The best solution found
```

The mathematical formulation of FA can be represented as follows:

1) Attractiveness: The attractiveness of a firefly is proportional to its brightness and decreases with distance:

$$\beta(r) = \beta_0e^{-\gamma r^2}$$

where β_0 is the attractiveness at distance $r = 0$,(29) γ is the light absorption coefficient, and r is the distance between two fireflies.

2) Movement: A firefly i is attracted to a brighter firefly j and moves according to:

$$x_i^{t+1} = x_i^t + \beta(r_{ij})(x_j^t - x_i^t) + \alpha\epsilon_i^t$$

where α is the randomization parameter, and ϵ_i^t(30) is a vector of random numbers drawn from a Gaussian distribution or uniform distribution.

E. Bacterial Foraging Optimization (BFO)

Bacterial Foraging Optimization is a bio-inspired optimization algorithm based on the foraging behavior of E. coli bacteria. Bacteria seek nutrients and avoid harmful substances by using mechanisms such as chemotaxis, swarming, reproduction, and elimination-dispersal.

The basic steps of BFO are as follows:

```
Algorithm 5: Bacterial Foraging Optimization
Initialize a population of N bacteria with random positions
for l = 1 to Ned (elimination-dispersal loops) do
    for k = 1 to Nre (reproduction loops) do
        for j = 1 to Nc (chemotaxis loops) do
            for i = 1 to N (bacteria) do
                Compute fitness of bacterium i
                Perform tumble (generate random direction)
                Perform swim (continue in same direction if improving)
            end for
        end for
        Sort bacteria based on accumulated fitness
        Perform reproduction (better half replicate, worse half eliminated)
    end for
    Perform elimination-dispersal (random replacement with probability p)
end for
return The best solution found
```

The mathematical formulation of BFO can be represented as follows:

1) Chemotaxis: Bacteria move by tumbling (changing direction) and swimming (moving in the same direction). The position update during chemotaxis is:

$$\theta^i(j+1, k, l) = \theta^i(j, k, l) + C(i)\frac{\Delta(i)}{\sqrt{\Delta^T(i)\Delta(i)}}$$

where $\theta^i(j, k, l)$ is the position of bacterium i at(31) chemotaxis step j, reproduction step k, and elimination-dispersal step l, $C(i)$ is the step size, and $\Delta(i)$ is a random direction vector.

2) Swarming: Bacteria can form groups and move as a unit. The cell-to-cell attraction and repulsion is modeled as:

$$J_{cc}(\theta, P(j,k,l)) = \sum_{i=1}^{S}\left[-d_{attract}\exp\left(-w_{attract}\sum_{m=1}^{p}(\theta_m - \theta_m^i)^2\right)\right] + \sum_{i=1}^{S}\left[h_{repellant}\exp\left(-w_{repellant}\sum_{m=1}^{p}(\theta_m - \theta_m^i)^2\right)\right]$$

where $d_{attract}$, $w_{attract}$, $h_{repellant}$, and $w_{repellant}$(32) are coefficients for the attraction and repulsion.

3) Reproduction: After Nc chemotaxis steps, bacteria are sorted based on their accumulated fitness, and the least healthy half die while the healthiest half split into two identical bacteria.

4) Elimination-Dispersal: With probability p_{ed}, a bacterium is eliminated and a new one is randomly dispersed in the search space.

V. AI-ENHANCED OPTIMIZATION FRAMEWORKS

A. Machine Learning Integration

We propose novel hybrid frameworks that integrate machine learning techniques with metaheuristic optimization algorithms to enhance their performance in solving multi-objective OPF problems with FACTS devices. These frameworks leverage the predictive capabilities of machine learning to guide the search process, improve convergence, and enhance solution quality.

1) Neural Network-Enhanced Solution Evaluation:

In traditional metaheuristic algorithms, evaluating each candidate solution requires solving the power flow equations, which is computationally expensive. We propose a neural network approach to approximate the power flow solutions, significantly reducing computational complexity. The neural network is trained on a dataset of input-output pairs, where inputs are control variables and outputs are power flow solutions.

The neural network architecture consists of an input layer with nodes corresponding to control variables, hidden layers with activation functions to capture non-linearities, and an output layer representing the power flow solution variables. The training process minimizes the mean squared error between the neural network predictions and actual power flow solutions:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

where y_i is the actual power flow solution, $\hat{y}_i$ is(33) the neural network prediction, and N is the number of training samples.

2) Meta-Model Based Optimization:

We employ a meta-model approach that uses machine learning to construct surrogate models of the objective functions and constraints. These surrogate models are then used within the optimization algorithms to guide the search towards promising regions of the solution space.

The surrogate model is updated iteratively using active learning techniques to balance exploration and exploitation. The acquisition function for selecting new points for evaluation is based on the expected improvement (EI):

$$EI(x) = (f_{min} - \hat{f}(x))\Phi\left(\frac{f_{min} - \hat{f}(x)}{s(x)}\right) + s(x)\phi\left(\frac{f_{min} - \hat{f}(x)}{s(x)}\right)$$

where f_{min} is the current minimum of the(34) objective function, $\hat{f}(x)$ is the predicted value at point x, $s(x)$ is the standard deviation of the prediction, Φ is the cumulative distribution function of the standard normal distribution, and ϕ is the probability density function of the standard normal distribution.

B. Deep Learning for Parameter Adaptation

We propose a deep learning approach for adaptive parameter setting in metaheuristic algorithms. The key parameters of each algorithm (e.g., crossover and mutation rates in GA, inertia weight and acceleration coefficients in PSO) significantly impact performance but are typically set empirically or tuned manually.

Our approach employs a deep neural network to predict optimal parameter values based on the current state of the optimization process. The network takes as input features describing the current population diversity, convergence rate, and problem characteristics, and outputs recommended parameter values.

The deep neural network architecture consists of multiple hidden layers with ReLU activation functions and a final output layer with sigmoid activation to bound the parameters within their valid ranges. The network is trained using a reinforcement learning approach, where the reward is based on the improvement in objective function value achieved with the recommended parameters:

$$R_t = \frac{f_{best,t-1} - f_{best,t}}{f_{best,t-1}}$$

where $f_{best,t}$ is the best objective function value(35) at iteration t.

C. Hybrid AI-Metaheuristic Frameworks

We develop novel hybrid frameworks that integrate AI techniques with metaheuristic algorithms in a synergistic manner. These frameworks leverage the complementary strengths of different approaches to enhance performance, robustness, and adaptability.

1) Neural-Enhanced Genetic Algorithm (NEGA):

The NEGA combines the evolutionary search capabilities of GA with the learning capabilities of neural networks. A recurrent neural network (RNN) is trained to predict promising search directions based on the evolutionary history. The movement operator in GA is modified to incorporate these predictions:

$$x_{new} = x_{old} + \alpha\Delta x_{GA} + (1 - \alpha)\Delta x_{NN}$$

where Δx_{GA} is the change suggested by the(36) standard GA operators, Δx_{NN} is the change suggested by the neural network, and α is a mixing parameter that balances the two components.

2) Deep Reinforcement Learning-Particle Swarm Optimization (DRL-PSO):

The DRL-PSO framework uses deep reinforcement learning to guide the movement of particles in PSO. A deep Q-network (DQN) is trained to predict the value of different movement directions. The velocity update in PSO is modified to incorporate these predictions:

$$v_i^{t+1} = wv_i^t + c_1r_1(pbest_i - x_i^t) + c_2r_2(gbest - x_i^t) + c_3r_3(qbest_i - x_i^t)$$

where $qbest_i$ is the direction with the highest Q-(37) value as predicted by the DQN, c_3 is a coefficient controlling the influence of the DQN, and r_3 is a random number in the range $[0, 1]$.

3) Adaptive Neuro-Fuzzy Firefly Algorithm (ANFFA):

The ANFFA integrates an adaptive neuro-fuzzy inference system (ANFIS) with the firefly algorithm to adaptively adjust the algorithm parameters based on the optimization state. The ANFIS takes as input measures of population diversity, convergence rate, and solution quality, and outputs adjusted values for the light absorption coefficient γ and the randomization parameter α:

$$\gamma_t = ANFIS_\gamma(diversity_t, convergence_t, quality_t)$$
$$\alpha_t = ANFIS_\alpha(diversity_t, convergence_t, quality_t)$$

These adaptive parameters enhance the(38) algorithm's ability to balance exploration and exploitation based on the current optimization state.

VI. TEST SYSTEMS AND SIMULATION METHODOLOGY

A. IEEE Test Systems

We evaluate the proposed algorithms and frameworks on five standard IEEE test systems of varying complexity:

1) IEEE 9-Bus System: This system consists of 3 generators, 9 buses, and 9 transmission lines. It serves as a simple test case for algorithm validation and preliminary performance assessment.

IEEE 9-Bus System

Fig. 3. IEEE 9-Bus test system.

2) IEEE 14-Bus System: This system represents a portion of the American Electric Power System and consists of 5 generators, 14 buses, and 20 transmission lines. It provides a moderately complex test case for evaluating algorithm performance.

3) IEEE 30-Bus System: This system is a simplified representation of the American Electric Power System and consists of 6 generators, 30 buses, and 41 transmission lines. It serves as a medium-sized test case for comprehensive algorithm evaluation.

4) IEEE 57-Bus System: This system consists of 7 generators, 57 buses, and 80 transmission lines. It provides a more challenging test case for assessing algorithm scalability and performance under increased complexity.

5) IEEE 118-Bus System: This system consists of 54 generators, 118 buses, and 186 transmission lines. It serves as a large-scale test case for evaluating algorithm performance on realistically sized power systems.

TABLE I: Characteristics of IEEE Test Systems

Test System	Buses	Generators	Lines	Loads	Base MVA
IEEE 9-Bus	9	3	9	3	100
IEEE 14-Bus	14	5	20	11	100
IEEE 30-Bus	30	6	41	21	100
IEEE 57-Bus	57	7	80	42	100
IEEE 118-Bus	118	54	186	91	100

B. Renewable Energy Integration

We integrate renewable energy sources into the test systems to evaluate the effectiveness of the proposed approaches in addressing the challenges associated with renewable energy variability and uncertainty. Wind and solar power plants are modeled as follows:

1) Wind Power Model: The power output of a wind turbine is modeled as a function of wind speed using the following expression:

$$P_{wind} = \begin{cases} 0, & v < v_{in} \text{ or } v > v_{out} \\ P_{rated}\cdot\frac{v^3 - v_{in}^3}{v_{rated}^3 - v_{in}^3}, & v_{in} \le v < v_{rated} \\ P_{rated}, & v_{rated} \le v \le v_{out} \end{cases}$$

where v is the wind speed, v_{in} is the cut-in wind(39) speed, v_{rated} is the rated wind speed, v_{out} is the cut-out wind speed, and P_{rated} is the rated power output of the wind turbine.

2) Solar Power Model: The power output of a solar photovoltaic (PV) panel is modeled as a function of solar irradiance and temperature:

$$P_{solar} = P_{STC}\cdot\frac{G}{G_{STC}}\cdot[1 - \gamma(T_c - T_{STC})]$$

where P_{STC} is the rated power output at(40) standard test conditions (STC), G is the solar irradiance, G_{STC} is the solar irradiance at STC (1000 W/m²), γ is the temperature coefficient of power, T_c is the cell temperature, and T_{STC} is the cell temperature at STC (25°C).

The renewable energy sources are integrated into the test systems at different penetration levels, ranging from 10% to 50% of the total generation capacity, to evaluate the robustness and adaptability of the proposed approaches under varying renewable energy conditions.

C. FACTS Device Placement Scenarios

We consider various scenarios for FACTS device placement to evaluate the effectiveness of the proposed approaches in optimizing power system performance. These scenarios include:

1) Single FACTS Device: A single FACTS device (SVC, TCSC, UPFC, or DPFC) is placed at the optimal location determined by the optimization algorithm.

2) Multiple Identical FACTS Devices: Multiple FACTS devices of the same type are placed at optimal locations determined by the optimization algorithm.

3) Multiple Different FACTS Devices: Multiple FACTS devices of different types are placed at optimal locations determined by the optimization algorithm.

4) Coordinated FACTS Device Placement and Control: The placement and control settings of multiple FACTS devices are optimized simultaneously to achieve the best overall system performance.

D. Performance Metrics

We evaluate the performance of the proposed algorithms and frameworks using the following metrics:

1) Objective Function Value: The value of each objective function and the weighted sum (if applicable) for the best solution found by each algorithm.

2) Convergence Rate: The number of iterations or function evaluations required to reach a specified level of solution quality.

3) Solution Quality: The diversity and spread of the Pareto front for multi-objective optimization, or the proximity to the global optimum for single-objective optimization.

4) Computational Efficiency: The computational time required to find high-quality solutions.

5) Algorithm Robustness: The consistency of performance across multiple runs with different random initializations.

E. Simulation Setup

All simulations are conducted using MATLAB software on a computer with an Intel Core i9 processor, 32 GB RAM, and NVIDIA GeForce RTX 3080 GPU. The power flow calculations are performed using MATPOWER, and the FACTS device models are implemented as custom extensions.

Each algorithm is run 30 times with different random initializations to ensure statistical significance of the results. The algorithm parameters are set based on preliminary tuning and literature recommendations. For the AI-enhanced frameworks, the neural networks are implemented using TensorFlow and Keras, with training conducted on a separate dataset of power flow solutions.

VII. RESULTS AND DISCUSSION

A. Comparative Analysis of Bio-Inspired Algorithms

We first compare the performance of the five bio-inspired metaheuristic algorithms (GA, PSO, ACO, FA, and BFO) in solving the multi-objective OPF problem without FACTS devices. Table II presents the best, worst, and average values of the weighted sum objective function for each algorithm on the IEEE 30-bus system, along with the standard deviation across 30 runs.

TABLE II: Comparison of Bio-Inspired Algorithms on IEEE 30-Bus System (Without FACTS)

Algorithm	Best Value	Worst Value	Average Value	Standard Deviation	Average Convergence Time (s)
GA	803.42	823.17	810.25	5.73	35.62
PSO	801.85	815.62	806.21	4.12	28.45
ACO	805.37	828.94	814.76	6.85	42.18
FA	800.93	812.41	804.75	3.47	31.76
BFO	802.76	824.59	811.84	6.23	45.91

The results in Table II indicate that FA achieves the best performance in terms of solution quality, followed by PSO. The FA also demonstrates the lowest standard deviation, indicating higher consistency across multiple runs. The PSO algorithm shows the fastest convergence, requiring significantly less computational time than the other algorithms.

Convergence Characteristics

Fig. 4. Convergence characteristics of bio-inspired algorithms on IEEE 30-bus system.

The convergence characteristics shown in Fig. 4 further illustrate the behavior of each algorithm. The PSO and FA algorithms demonstrate faster initial convergence, while the GA and BFO algorithms show more gradual improvement throughout the iterations. The ACO algorithm exhibits slower convergence but continues to improve over a larger number of iterations.

B. FACTS Device Placement and Sizing Results

We next evaluate the effectiveness of the bio-inspired algorithms in determining the optimal location and sizing of FACTS devices. Table III presents the results for the IEEE 30-bus system with a single SVC device, showing the optimal location, size, and resulting improvement in various performance metrics.

TABLE III: Optimal SVC Placement Results for IEEE 30-Bus System

Algorithm	Optimal Bus	SVC Size (MVAr)	Cost Reduction (%)	Loss Reduction (%)	Voltage Deviation Reduction (%)
GA	24	35.8	2.45	8.72	12.36
PSO	24	37.2	2.58	9.15	13.27
ACO	26	33.5	2.32	8.41	11.83
FA	24	38.4	2.67	9.42	13.85
BFO	26	34.7	2.39	8.58	12.14

The results in Table III show that the FA algorithm identifies the most effective SVC placement, leading to the highest reductions in generation cost, transmission losses, and voltage deviation. Most algorithms identify bus 24 as the optimal location for the SVC, with size ranging from 33.5 to 38.4 MVAr. The improvements demonstrate the significant impact of optimally placed FACTS devices on power system performance.

Voltage Profile

Fig. 5. Voltage profile of IEEE 30-bus system before and after SVC placement using FA.

Fig. 5 illustrates the voltage profile improvement achieved through optimal SVC placement using the FA algorithm. The voltage magnitudes at various buses are brought closer to their reference values, leading to a more uniform voltage profile and reduced voltage deviation across the system.

C. Multi-Objective Optimization Results

For the multi-objective optimization, we employ the Pareto optimization approach to identify the set of non-dominated solutions that represent different trade-offs between the objectives. Fig. 6 shows the Pareto front obtained by the FA algorithm for the IEEE 30-bus system with a single UPFC device, considering the objectives of generation cost minimization and transmission loss minimization.

Pareto Front

Fig. 6. Pareto front for generation cost minimization and transmission loss minimization obtained by FA algorithm.

The Pareto front illustrates the trade-off between generation cost and transmission losses. Solutions at the lower left of the front represent balanced trade-offs, while solutions at the extremes prioritize one objective over the other. Decision-makers can select the most appropriate solution based on their specific requirements and preferences.

Table IV presents three representative solutions from the Pareto front, highlighting the trade-offs between different objectives and the corresponding UPFC parameters.

TABLE IV: Representative Solutions from Pareto Front (IEEE 30-Bus System with UPFC)

Solution	Generation Cost ($/h)	Losses (MW)	UPFC Location (From-To Bus)	Series Voltage Magnitude (p.u.)	Series Voltage Angle (deg)
Solution A (Cost Priority)	798.42	9.24	10-20	0.075	85.3
Solution B (Balanced)	802.76	8.13	10-20	0.082	78.6
Solution C (Loss Priority)	814.35	7.48	6-10	0.095	72.1

D. AI-Enhanced Framework Performance

We evaluate the performance of the proposed AI-enhanced optimization frameworks compared to the standard bio-inspired algorithms. Table V presents the results for the IEEE 118-bus system with multiple FACTS devices, showing the improvement achieved by each AI-enhanced framework over its corresponding standard algorithm.

TABLE V: Performance Improvement of AI-Enhanced Frameworks (IEEE 118-Bus System with Multiple FACTS Devices)

Framework	Objective Function Improvement (%)	Convergence Time Reduction (%)	Standard Deviation Reduction (%)
NEGA (vs. GA)	8.23	42.15	35.67
DRL-PSO (vs. PSO)	6.78	38.94	31.42
ANFFA (vs. FA)	5.46	35.28	27.85

The results in Table V demonstrate the significant improvements achieved by the AI-enhanced frameworks. The Neural-Enhanced Genetic Algorithm (NEGA) shows the most substantial improvement, with an 8.23% reduction in the objective function value, a 42.15% reduction in convergence time, and a 35.67% reduction in standard deviation. These improvements highlight the effectiveness of integrating AI techniques with metaheuristic algorithms for solving complex power system optimization problems.

Convergence Comparison

Fig. 7. Convergence comparison between standard algorithms and AI-enhanced frameworks on IEEE 118-bus system.

Fig. 7 illustrates the convergence characteristics of the standard algorithms and their AI-enhanced counterparts. The AI-enhanced frameworks demonstrate significantly faster convergence and achieve better final solutions, highlighting the benefits of the integration.

E. Impact of Renewable Energy Integration

We assess the effectiveness of the proposed approaches in addressing the challenges associated with renewable energy integration. Table VI presents the results for different renewable energy penetration levels on the IEEE 30-bus system, showing the performance of the ANFFA algorithm with and without FACTS devices.

TABLE VI: Impact of Renewable Energy Penetration (IEEE 30-Bus System, ANFFA Algorithm)

Renewable Penetration (%)	Generation Cost ($/h) Without FACTS	Generation Cost ($/h) With FACTS	Losses (MW) Without FACTS	Losses (MW) With FACTS	Improvement (%)
10	792.34	784.15	8.76	7.94	9.36
20	785.67	774.28	9.14	8.12	11.16
30	778.92	763.45	9.82	8.43	14.15
40	772.18	752.67	10.56	8.91	15.63
50	765.43	742.18	11.75	9.53	18.89

The results in Table VI show that as renewable energy penetration increases, the improvements achieved through FACTS devices become more significant. At 50% renewable penetration, the FACTS devices provide an 18.89% improvement in the combined objective, highlighting their crucial role in facilitating high renewable energy integration.

Renewable Energy Impact

Fig. 8. Impact of renewable energy penetration on system performance with and without FACTS devices.

Fig. 8 illustrates the impact of increasing renewable energy penetration on system performance. Without FACTS devices, the system experiences increasing challenges in terms of voltage deviations and losses as renewable penetration increases. With optimally placed and controlled FACTS devices, these challenges are effectively mitigated, enabling higher renewable energy integration while maintaining system performance.

F. Scalability Analysis

We evaluate the scalability of the proposed approaches by applying them to test systems of increasing size. Table VII presents the computation time and solution quality for different algorithms on various test systems.

TABLE VII: Scalability Analysis of Algorithms Across Different Test Systems

Test System	Algorithm	Computation Time (s)	Relative Solution Quality (%)
IEEE 9-Bus	FA	12.45	100.00
	PSO	10.82	98.73
	ANFFA	15.31	101.86
IEEE 30-Bus	FA	28.76	100.00
	PSO	25.42	97.85
	ANFFA	32.15	103.42
IEEE 57-Bus	FA	58.34	100.00
	PSO	52.17	96.28
	ANFFA	64.92	104.75
IEEE 118-Bus	FA	142.68	100.00
	PSO	128.35	94.76
	ANFFA	128.43	107.28

The results in Table VII show that while the standard FA algorithm maintains consistent relative performance across different system sizes, the ANFFA framework's advantage grows with system size. For the IEEE 118-bus system, ANFFA achieves a 7.28% improvement in solution quality compared to FA, despite requiring less computational time than FA for larger systems. This demonstrates the scalability and efficiency of the AI-enhanced framework for large-scale power system optimization problems.

VIII. CONCLUSION

This paper has presented a comprehensive analytical framework for evaluating and enhancing bio-inspired metaheuristic optimization techniques for solving multi-objective OPF problems with FACTS device placement and control in power systems with renewable energy integration. The main contributions and findings of this research are summarized as follows:

1) Among the five bio-inspired algorithms evaluated (GA, PSO, ACO, FA, and BFO), the Firefly Algorithm (FA) demonstrated the best overall performance in terms of solution quality and consistency across different test systems and problem formulations.

2) The optimal placement and sizing of FACTS devices using bio-inspired algorithms resulted in significant improvements in power system performance, including reduced generation costs, transmission losses, and voltage deviations. The effectiveness of these improvements increased with higher renewable energy penetration, highlighting the crucial role of FACTS devices in facilitating renewable energy integration.

3) The proposed AI-enhanced optimization frameworks, particularly the Neural-Enhanced Genetic Algorithm (NEGA) and Adaptive Neuro-Fuzzy Firefly Algorithm (ANFFA), achieved substantial improvements over their standard counterparts in terms of solution quality, convergence speed, and consistency. These improvements were more pronounced for larger and more complex systems, demonstrating the scalability and efficiency of the AI-enhanced approaches.

4) The multi-objective optimization approach based on Pareto optimization provided a comprehensive set of trade-off solutions, allowing decision-makers to select the most appropriate solution based on their specific requirements and preferences. The identified Pareto-optimal solutions demonstrated different trade-offs between generation cost, transmission losses, voltage profile improvement, and stability enhancement.

5) The integration of renewable energy sources introduced additional challenges in terms of variability and uncertainty, which were effectively addressed through the optimal placement and control of FACTS devices. The proposed approaches demonstrated robust performance under different renewable energy penetration levels, enabling higher renewable integration while maintaining system performance.

Future research directions include the development of more advanced AI-enhanced optimization frameworks that can better handle the uncertainties associated with renewable energy generation, the extension of the proposed approaches to consider the dynamic behavior of power systems, and the exploration of distributed optimization techniques for large-scale power system applications.

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